PHYS 350: Honours Electricity and Magnetism

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Abstract

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1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

August 31 - September 4, 2026

2.1 Vector Algebra

Definition 1 (Scalar)**.** *A scalar is a quantity with just a magnitude.*

Definition 2 (Vector)**.** *A vector is a quantity with a magnitude and a direction, i.e.* $\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z}$.

Definition 3 (Cross Product)**.** *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors:*

$$\vec{v}\times\vec{w} = |\vec{v}||\vec{w}|\sin\theta\,\hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \hat{x}(v_yw_z - w_yv_z) - \hat{y}(v_xw_z - w_xv_z) + \hat{z}(v_xw_y - w_xv_y). \tag{1}$$

Definition 4 (Dot Product)**.** *The dot product measures how much of* $\vec{v}$ *points along* $\vec{w}$*, so it is zero for perpendicular vectors:*

$$\vec{v}\cdot\vec{w} = v_xw_x + v_yw_y + v_zw_z = |\vec{v}||\vec{w}|\cos\theta. \tag{2}$$

Remark 1. *Note* $\vec{v}\times\vec{w} = -\vec{w}\times\vec{v}$*, and* $|\vec{v}| = \sqrt{\vec{v}\cdot\vec{v}}$.

Definition 5 (Field)**.** *A field is something with a value at every point in space. A scalar field assigns a number to each point. A vector field assigns a vector to each point.*

2.2 Differential Calculus

Definition 6 (Del operator)**.** *The del operator*

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right).$$

Definition 7 (Gradient)**.** *The gradient turns a scalar into a vector, pointing in the direction of steepest increase of* f*:*

$$\vec{\nabla}f = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z}\hat{z}. \tag{3}$$

Remark 2. *The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f.*

Definition 8 (Divergence)**.** *Divergence turns a vector into a scalar, measuring sources and sinks. For a small box, non-zero divergence means the inflow and outflow through its faces do not balance: arrows spreading from a point (positive) mean fluid is created there, arrows converging (negative) mean it is destroyed.*

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \tag{4}$$

Definition 9 (Curl)**.** *Curl turns a vector into a vector, measuring local rotation. Picture a small paddle wheel: if the flow spins it, the curl is non-zero there. Uniform flow does not spin it, but shear flow (parallel arrows, longer on one side) does, so straight field lines do not imply zero curl.*

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial}{\partial y} F_z - \frac{\partial}{\partial z} F_y, -(\frac{\partial}{\partial x} F_z - \frac{\partial}{\partial z} F_x), \frac{\partial}{\partial x} F_y - \frac{\partial}{\partial y} F_x \right). \tag{5}$$

Remark 3. *The direction of the curl follows the right-hand rule: curl your fingers along the paddle wheel's rotation and your thumb points along $\vec{\nabla} \times \vec{F}$.*

Remark 4. *Divergence and curl are themselves fields: they can vanish in some places and not others, and a general field has both.*

2.3 Integral Calculus

Definition 10 (Line Integral)**.** *A line integral sums $\vec{F}$ along a curve, picking up only the component tangent to the path:*

$$\int_C \vec{F} \cdot d\vec{l}.$$

Example. *Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R.*

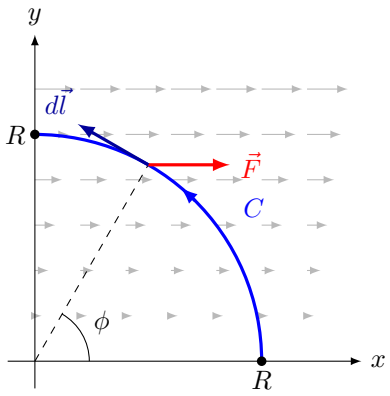


Figure 1: The field $\vec{F} = y\hat{x}$ (gray) and the quarter circle C. At a typical point, $\vec{F}$ points along $+\hat{x}$ while $d\vec{l}$ has a $-\hat{x}$ component, so $\vec{F} \cdot d\vec{l} < 0$.

On the circle $d\vec{l} = \frac{d\vec{r}}{d\phi} d\phi = R(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y})d\phi$ *and* $y = R\sin\phi$, *so*

$$\int_C \vec{F} \cdot d\vec{l} = \int_0^{\pi/2} \begin{pmatrix} R\sin\phi \\ 0 \end{pmatrix} \cdot R \begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix} d\phi = -R^2 \int_0^{\pi/2} \sin^2\phi\, d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ *points along* $+\hat{x}$ *while we travel in the* $-\hat{x}$ *direction over most of the arc.*

Definition 11 (Flux)**.** *Flux measures how much of the field pokes through a surface:*

$$\int_S \vec{F} \cdot d\vec{a} = \int_S \vec{F} \cdot \hat{n}\, da.$$

Remark 5. *Flux is the total amount of a vector field flowing through a surface. Divergence is the flux per unit volume at a specific point.*

Definition 12 (Volume Integral)**.** *A volume integral sums a scalar over a region:*

$$\int_V f\, d^3\vec{r} = \iiint_V f\, dx\, dy\, dz.$$

Remark 6. *We write* $\oint$ *when the curve is a closed loop or the surface is closed.*

2.3.1 The Fundamental Theorems

Theorem 1 (The Fundamental Theorem of Calculus)**.** *The boundary of an interval is two points:*

$$\int_a^b f'(x)\, dx = f(b) - f(a).$$

Theorem 2 (The Fundamental Theorem for Gradients)**.**

$$\int_a^b \vec{\nabla} T \cdot d\vec{l} = T(b) - T(a).$$

The boundary of a curve is its two endpoints. To find the Eiffel Tower's height, you can sum the rise of every step (left side) or subtract altimeter readings at the top and bottom (right side).

Theorem 3 (The Fundamental Theorem for Curls)**.** *Another name for this theorem is Stokes' theorem.*

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{l}.$$

Geometric meaning: to get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside. Neighbouring whirlpools cancel along their shared edges, leaving only the boundary.

Theorem 4 (The Fundamental Theorem for Divergence)**.** *It has 2 important names: Gauss's theorem (or the divergence theorem) and Green's theorem. Respectively:*

$$\oiint_S (\vec{F} \cdot \hat{n}) dS = \iiint_V (\vec{\nabla} \cdot \vec{F}) dV,$$

$$\oint_C P dx + Q dy = \iint_R (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}) dx dy.$$

2.4 Spherical Coordinates

Definition 13 (Spherical coordinates)**.** *r is the distance from the origin, $\theta \in [0, \pi]$ the angle down from the $+z$ axis, and $\phi \in [0, 2\pi)$ the angle around it. The unit vectors $\hat{r}, \hat{\theta}, \hat{\phi}$ change direction from point to point, so they cannot be pulled out of an integral; convert to $\hat{x}, \hat{y}, \hat{z}$ first. The elements are*

$$d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}, \qquad d\tau = r^2 \sin\theta\,dr\,d\theta\,d\phi, \qquad d\vec{a}_{sphere} = R^2 \sin\theta\,d\theta\,d\phi\,\hat{r}.$$

Proposition 1 (Derivatives in spherical coordinates)**.**

$$\vec{\nabla} f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\hat{\phi},$$

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 F_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\, F_\theta\right) + \frac{1}{r\sin\theta}\frac{\partial F_\phi}{\partial \phi},$$

$$\vec{\nabla} \times \vec{F} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, F_\phi) - \frac{\partial F_\theta}{\partial \phi}\right]\hat{r} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial F_r}{\partial \phi} - \frac{\partial}{\partial r}(r F_\phi)\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r F_\theta) - \frac{\partial F_r}{\partial \theta}\right]\hat{\phi},$$

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}.$$

Remark 7. *For a purely radial field $\vec{F} = F_r(r)\hat{r}$, only the first term of the divergence survives, and the curl vanishes. So every central field $f(r)\hat{r}$ is curl-free.*

2.5 Index Notation

Remark 8. *A matrix carries two indices, A_{ij}, where i says which row and j says which column.*

Definition 14 (Transpose)**.** *The transpose swaps the indices, $(A^T)_{ij} = A_{ji}$.*

Definition 15 (Symmetry)**.** *A symmetric matrix has $A_{ij} = A_{ji}$, and an anti-symmetric one has $B_{ij} = -B_{ji}$.*

Definition 16 (Kronecker delta)**.** *The Kronecker delta equals $+1$ when its condition is fulfilled and 0 otherwise, so it could be the identity matrix in index notation:*

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Example.

$$\sum_j \delta_{ij} v_j = v_i.$$

Proposition 2 (Matrix Multiplication)**.**

$$\vec{w} = A\vec{v} \implies w_i = \sum_j A_{ij} v_j.$$

Proposition 3 (Dot Product)**.**

$$\vec{v} \cdot \vec{w} = \sum_i v_i w_i; \vec{\nabla} \cdot \vec{F} = \sum_i \partial_i F_i.$$

Proposition 4 (Cross Product)**.**

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k; (\vec{\nabla} \times \vec{F})_i = \sum_{j,k} \varepsilon_{ijk} \partial_j F_k.$$

Definition 17 (Levi-Civita symbol)**.** *ε_{ijk} is defined by*

- $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = +1$.
- $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$.
- *If any two indices are equal, $\varepsilon_{ijk} = 0$.*

Example. *For $i = 1$, only the terms with $(j, k) = (2, 3)$ and $(3, 2)$ survive:*

$$(\vec{v} \times \vec{w})_1 = \varepsilon_{123} v_2 w_3 + \varepsilon_{132} v_3 w_2 = v_2 w_3 - v_3 w_2.$$

Proposition 5 (Epsilon-delta identity)**.** *Contracting two Levi-Civita symbols on a shared index gives*

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}.$$

2.6 Second Derivatives Identities

Definition 18 (Laplacian)**.** *Taking the divergence of a gradient gives*

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f, \tag{6}$$

where ∇^2 *is the Laplacian*[1]*.*

Remark 9. ∇^2 *also acts on vector fields, component by component:* $(\nabla^2 \vec{F})_i = \nabla^2 F_i$*.*

The full list of second derivative identities:

Name	**Identity**
Divergence of a gradient	$\vec{\nabla} \cdot (\vec{\nabla} f) = \nabla^2 f$
Curl of a gradient	$\vec{\nabla} \times (\vec{\nabla} f) = 0$
Divergence of a curl	$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$
Curl of a curl	$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$

Table 1: Second derivative identities.

The curl of a curl identity and the divergence of a gradient identity are proved in Appendix A.2.

2.7 The Dirac Delta Function

Definition 19 (Dirac delta)**.** $\delta(x)$ *is zero everywhere except at* $x = 0$*, where it is an infinitely tall spike of unit area:*

$$\int_{-\infty}^{\infty} \delta(x)\, dx = 1.$$

Proposition 6 (Properties)**.** *(i) Sifting:* $\int_a^b f(x)\, \delta(x - x_0)\, dx = f(x_0)$ *if* $a < x_0 < b$*, and* 0 *if* x_0 *is outside* $[a, b]$*.*

(ii) Scaling: $\delta(kx) = \frac{1}{|k|}\delta(x)$*.*

(iii) In 3D: $\int_V f(\vec{r})\, \delta^3(\vec{r} - \vec{r}_0)\, d\tau = f(\vec{r}_0)$ *if* $\vec{r}_0 \in V$*.*

Example. $\int_0^3 x^3 \delta(x + 1)\, dx = 0$ *since* $x = -1$ *is outside the range, while*

$$\int_{-1}^{1} 9x^2\, \delta(3x + 1)\, dx = \frac{1}{3} \int_{-1}^{1} 9x^2\, \delta\big(x + \tfrac{1}{3}\big)\, dx = \frac{1}{3} \cdot 9 \cdot \frac{1}{9} = \frac{1}{3}.$$

[1] Mathematicians often write Δ, but in physics we use ∇^2.

Theorem 5 (Divergence of $\hat{r}/r^2$)**.** *By the spherical divergence,* $\vec{\nabla}\cdot\left(\hat{r}/r^2\right) = \frac{1}{r^2}\partial_r(1) = 0$ *for* $r \neq 0$*, yet its flux through any sphere is* $\oint \frac{1}{R^2} R^2 \sin\theta \, d\theta \, d\phi = 4\pi$*. By the divergence theorem, everything sits at the origin:*

$$\vec{\nabla}\cdot\left(\frac{\hat{r}}{r^2}\right) = 4\pi\,\delta^3(\vec{r}).$$

So for a point charge, $\vec{\nabla}\cdot\vec{E} = \frac{q}{\epsilon_0}\delta^3(\vec{r}) = \rho/\epsilon_0$*, as Gauss' law requires.*

2.8 The Theory of Vector Fields

Remark 10. *The condition at infinity is needed: a field can be both curl-free and divergence-free without being zero, e.g.* $\vec{F} = yz\,\hat{x} + zx\,\hat{y} + xy\,\hat{z} = \vec{\nabla}(xyz)$*. Adding it to any field changes neither* $\vec{\nabla}\cdot\vec{F}$ *nor* $\vec{\nabla}\times\vec{F}$*.*

Theorem 6 (Irrotational fields)**.** *Let* $\vec{F}$ *be a* C^1 *vector field on a simply connected domain (e.g. all of* $\mathbb{R}^3$*). The following are equivalent:*

(i) $\vec{\nabla}\times\vec{F} = 0$ *everywhere.*

(ii) $\int_a^b \vec{F}\cdot d\vec{l}$ *is independent of path, for any given end points.*

(iii) $\oint_C \vec{F}\cdot d\vec{l} = 0$ *for every closed loop* C*.*

(iv) $\vec{F} = -\vec{\nabla}\psi$ *for some scalar field* ψ*.*

Theorem 7 (Solenoidal fields)**.** *Let* $\vec{F}$ *be a* C^1 *vector field on a domain with no enclosed voids (e.g. all of* $\mathbb{R}^3$*). The following are equivalent:*

(i) $\vec{\nabla}\cdot\vec{F} = 0$ *everywhere.*

(ii) $\int_S \vec{F}\cdot d\vec{a}$ *is independent of surface, for any given boundary line.*

(iii) $\oint_S \vec{F}\cdot d\vec{a} = 0$ *for any closed surface* S*.*

(iv) $\vec{F} = \vec{\nabla}\times\vec{A}$ *for some vector field* $\vec{A}$*.*

Remark 11. *In electrostatics we will find* $\vec{\nabla}\times\vec{E} = 0$*, which licenses* $\vec{E} = -\vec{\nabla}V$ *and the electric potential. In magnetostatics we will find* $\vec{\nabla}\cdot\vec{B} = 0$*, which licenses* $\vec{B} = \vec{\nabla}\times\vec{A}$*, the vector potential.*

Theorem 8 (Helmholtz)**.** *If* $\vec{\nabla}\cdot\vec{F}$ *and* $\vec{\nabla}\times\vec{F}$ *are specified everywhere and* $\vec{F}\to 0$ *at infinity, then* $\vec{F}$ *is uniquely determined. It can be written as* $\vec{F} = \vec{\nabla}\psi + \vec{\nabla}\times\vec{A}$*, an irrotational part plus a solenoidal part.*

3 Electrostatics

September 11-16 & 21-25, 2026

Definition 20 (Electrostatics)**.** *Electrostatics is the study of stationary electric charges.*

3.1 Electric Field

Goal of electrodynamics: given source charges with known positions, find the force on a test charge Q.

Proposition 7 (Principle of Superposition)**.** *The force between two charges is unaffected by other charges, so*

$$\vec{F} = \sum_i \vec{F}_i.$$

Law 1 (Coulomb's law)**.** *The force on a test charge Q due to a point charge q is*

$$\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{qQ}{\mathfrak{r}^2}\,\hat{\mathfrak{r}}\,[N]\,, \tag{7}$$

where $\vec{\mathfrak{r}} = \vec{r} - \vec{r'}$ is the separation vector from the location of q ($\vec{r'}$) to Q ($\vec{r}$).

Remark 12. *Coulomb's law has the same shape as Newton's gravity. The reason is geometric rather than a coincidence, and Gauss' law makes that explicit: in d spatial dimensions the flux spreads over a $(d-1)$-dimensional sphere, so $F \propto 1/\mathfrak{r}^{d-1}$, and the 4π is the area of the unit sphere in $d = 3$.*

Definition 21 (Permittivity of free space)**.** *The permittivity of free space measures how easily an electric field can form in a classical vacuum:*

$$\epsilon_0 = 8.85 \times 10^{-12}\ \left[\frac{C^2}{N \cdot m^2}\right].$$

By the principle of superposition, the total force on a test charge Q due to several point charges is

$$\vec{F} = \frac{Q}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\mathfrak{r}_i^2}\,\hat{\mathfrak{r}}_i.$$

Which leads us to the definition of the electric field

$$\vec{F} = Q\vec{E}.$$

Definition 22 (Electric field)**.** *The electric field is the force per unit test charge.*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{\mathfrak{r}_i^2}\,\hat{\mathfrak{r}}_i. \tag{8}$$

$\vec{\mathfrak{r}}_i \equiv \vec{r} - \vec{r}_i$, where $\vec{r}$ is the location of the field point, where we want to know what force a charge would feel, and $\vec{r}_i$ is the location of the source charge q_i.

3.1.1 Continuous Charge Distributions

Definition 23 (Continuous distributions)**.** *Replacing q_i by $dq = \lambda\, dl' = \sigma\, da' = \rho\, d\tau'$ turns the sum into an integral:*

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{\mathfrak{r}^2}\,\hat{\mathfrak{r}}, \qquad \vec{\mathfrak{r}} = \vec{r} - \vec{r}', \tag{9}$$

where $\vec{r}$ is the field point and $\vec{r}'$ the source point.

Remark 13. *$\hat{\mathfrak{r}}$ is not constant: it points differently for every source point $\vec{r}'$, so it cannot come out of the integral. Split it into components first.*

Example (Line of charge)**.** *A straight segment of length $2L$, centred at the origin on the x axis, carries a uniform line charge density λ. Find $\vec{E}$ at the point $P = (0, 0, z)$, a height z above its midpoint.*

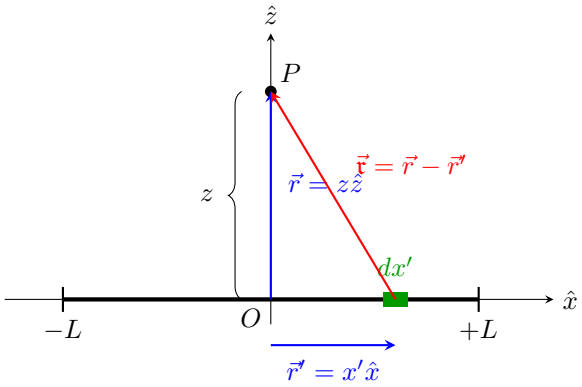


Figure 2: Setup for the finite line of charge. The source point $\vec{r}' = x'\hat{x}$ runs from $-L$ to L; the field point is P at $\vec{r} = z\hat{z}$; the separation vector is $\vec{\mathfrak{r}} = \vec{r} - \vec{r}'$.

With $\vec{r} = z\hat{z}$ and $\vec{r}' = x'\hat{x}$,

$$\vec{\mathfrak{r}} = z\hat{z} - x'\hat{x}, \qquad \mathfrak{r} = \sqrt{z^2 + x'^2},$$

so

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0}\int_{-L}^{L} \frac{z\hat{z} - x'\hat{x}}{(z^2 + x'^2)^{3/2}}\,dx' = \frac{\lambda}{4\pi\epsilon_0}\left[\hat{z}\int_{-L}^{L}\frac{z}{(z^2 + x'^2)^{3/2}}dx' - \hat{x}\int_{-L}^{L}\frac{x'}{(z^2 + x'^2)^{3/2}}dx'\right].$$

The second integral vanishes by symmetry: its integrand is odd in x', so the pieces at x' and $-x'$ cancel. The first is done with the substitution $x' = z\tan\theta$, giving

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0}\,\frac{1}{z}\left[\frac{x'}{\sqrt{x'^2 + z^2}}\right]_{-L}^{L}\hat{z} = \frac{\lambda}{4\pi\epsilon_0}\,\frac{2L}{z\sqrt{L^2 + z^2}}\,\hat{z} = \frac{\lambda}{2\pi\epsilon_0 z}\cdot\frac{L}{\sqrt{L^2 + z^2}}\,\hat{z}.$$

This has two limits:

- *Close to the line ($z \ll L$), it looks like an infinite line:* $\vec{E} \to \dfrac{\lambda}{2\pi\epsilon_0 z}\hat{z}$.
- *Far from the line ($z \gg L$), it looks like a point charge:* $\vec{E} \to \dfrac{Q}{4\pi\epsilon_0 z^2}\hat{z}$ *with* $Q = 2\lambda L$.

Remark 14. *Sanity check: an answer for an extended distribution should reduce to a simpler known field in the limits (up close and far away). If it does not, the answer is wrong.*

3.2 Gauss' Law

Law 2 (Gauss' law)**.** *The flux of $\vec{E}$ through any closed surface is set by the charge inside it:*

$$\oiint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}. \tag{10}$$

Where Q_{enc}/ϵ_0 comes from. Start with a single point charge q at the centre of a sphere of radius r. On the sphere $\vec{E}$ is radial and $d\vec{a} = \hat{r}\, da$, so by Coulomb's law

$$\oiint_S \vec{E} \cdot d\vec{a} = \frac{q}{4\pi\epsilon_0 r^2} \oiint_S da = \frac{q}{4\pi\epsilon_0 r^2} \cdot 4\pi r^2 = \frac{q}{\epsilon_0}.$$

The r^2 cancels, so every sphere around q gets the same flux. Three observations extend this:

- Any closed surface around q, of any shape: the same field lines that pierce the sphere must pierce it too, so its flux is still q/ϵ_0.
- A charge outside the surface: each of its field lines enters once and leaves once, so it contributes zero net flux.
- Many charges: by superposition (Principle 7) the fluxes add, and only the enclosed charges contribute, giving Q_{enc}/ϵ_0.

□

Proposition 8 (Differential form)**.** *Apply the divergence theorem (Theorem 4) to the left side and write $Q_{enc} = \iiint_V \rho\, d\tau$:*

$$\iiint_V (\vec{\nabla} \cdot \vec{E})\, d\tau = \frac{1}{\epsilon_0} \iiint_V \rho\, d\tau.$$

Since this holds for every volume V, the integrands match:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \tag{11}$$

Example. *The differential form reads ρ off a known field: $\vec{E} = Ax\,\hat{x}$ gives $\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E} = \epsilon_0 A$, a uniform charge density.*

Definition 24 (Gaussian Surface)**.** *The Gaussian surface just means an imaginary surface (nothing is physically there).*

Remark 15. *If there is no charge inside the Gaussian surface, the net flux is zero: every field line that enters on one side leaves on another.*

Remark 16 (Recipe)**.** *Gauss' law always holds, but it only solves for $\vec{E}$ when symmetry lets E come out of the integral, which means spherical, cylindrical or planar symmetry.*

1. *Use symmetry to fix the direction of $\vec{E}$ and what it depends on.*
2. *Pick a surface on which $\vec{E}$ is either parallel to $d\vec{a}$ with constant magnitude, or perpendicular to it.*
3. *Find Q_{enc}; for a non-uniform ρ, integrate it with the right volume element.*
4. *Solve $E \times (\text{area}) = Q_{enc}/\epsilon_0$, separately inside and outside the charge.*

A uniformly charged cube has no such surface, so its field outside is not that of a point charge at its centre (though it approaches it far away).

3.2.1 Spherical symmetry

Example (Point charge)**.** *Surround a point charge q with a sphere S of radius r centred on it. Two facts reduce the integral to a product:*

- *By symmetry, $\vec{E} = E(r)\,\hat{r}$: the field is radial and has the same magnitude everywhere on S.*
- *By geometry, $d\vec{a} = \hat{n}\,da = \hat{r}\,da$ on a sphere.*

So $\vec{E} \cdot d\vec{a} = E\,da$, and

$$\oiint_S \vec{E} \cdot d\vec{a} = E \underbrace{\oiint_S da}_{4\pi r^2} = \frac{q}{\epsilon_0} \quad \Longrightarrow \quad E = \frac{q}{4\pi\epsilon_0 r^2}.$$

Example (Uniform solid sphere)**.** *A ball of radius R and charge q. Inside, a sphere of radius r encloses $q\,r^3/R^3$, so*

$$\vec{E} = \begin{cases} \dfrac{q\,r}{4\pi\epsilon_0 R^3}\hat{r} & r < R \\ \dfrac{q}{4\pi\epsilon_0 r^2}\hat{r} & r > R \end{cases}.$$

E grows linearly inside and is continuous at R, since there is no surface charge. A thin shell instead gives $\vec{E} = 0$ inside and a jump of σ/ϵ_0 at R.

Remark 17. *With spherical symmetry, $E(r)$ depends only on the charge inside radius r. Charge further out does not change the field inside it, and the field outside only knows the total. So a uniformly charged shell that pulses in and out, $R(t) = R_0 + A\sin\omega t$, keeps the constant field $\frac{Q}{4\pi\epsilon_0 r^2}\hat{r}$ outside: accelerating charges, but nothing escapes to infinity, so no radiation.*

3.2.2 Cylindrical symmetry

Example (Infinite line of charge)**.** *This recovers the $z \ll L$ limit of the line of charge example in two lines. For a line with charge λ per unit length, take a coaxial[2] cylinder of radius z and length d. On each end cap (a flat disk), the arrows of $\vec{E}$ lie in the disk, pointing outward from its centre like the spokes of a wheel: perpendicular to the line of charge and parallel to the cap. So $\vec{E} \cdot d\vec{a} = 0$ on the caps and only the curved side counts:*

$$E\,(2\pi z d) = \frac{\lambda d}{\epsilon_0},$$

so

$$E = \frac{\lambda}{2\pi\epsilon_0 z}. \tag{12}$$

The field falls off as $1/z$, not $1/z^2$.

Example (Finite versus infinite line)**.** *Near its midpoint, a finite line has a weaker field than an infinite one with the same λ: the cylinder encloses the same charge, but some of the flux now leaks through the caps.*

Problem 1 (Midterm 2025, Q2)**.** *An infinite non-conducting tube along the x axis has $\rho(s) = A\,s/a$ for $s < a$ and 0 elsewhere, with s the distance from the axis. Find $\vec{E}$ everywhere.*

Solution 1. *By symmetry $\vec{E} = E(s)\hat{s}$, pointing straight away from the axis. Take a coaxial cylinder of radius s and length L; the caps carry no flux. With $d\tau = 2\pi s' L\,ds'$,*

$$Q_{enc} = \int_0^s A\frac{s'}{a}\,2\pi s' L\,ds' = \frac{2\pi A L\,s^3}{3a} \quad (s < a), \qquad Q_{enc} = \frac{2\pi A L\,a^2}{3} \quad (s > a).$$

Then $E\,(2\pi s L) = Q_{enc}/\epsilon_0$ gives

$$\vec{E} = \begin{cases} \dfrac{As^2}{3\epsilon_0 a}\hat{s} & s < a \\[2ex] \dfrac{Aa^2}{3\epsilon_0 s}\hat{s} & s > a \end{cases},$$

continuous at $s = a$, and falling as $1/s$ outside like a line charge.

3.2.3 Planar symmetry

Example (Infinite plane of charge)**.** *For a plane with surface charge density σ, use a Gaussian pillbox: a short flat cylinder (like a hockey puck) that pierces the plane, half above and half below. It has two flat caps of area A, one on each side of the plane and parallel to it, joined by a side wall. By symmetry $\vec{E}$ points*

[2] Sharing a common central axis.

straight away from the plane on both sides, so it runs along the side wall (no flux there) and straight through each cap (flux EA each):

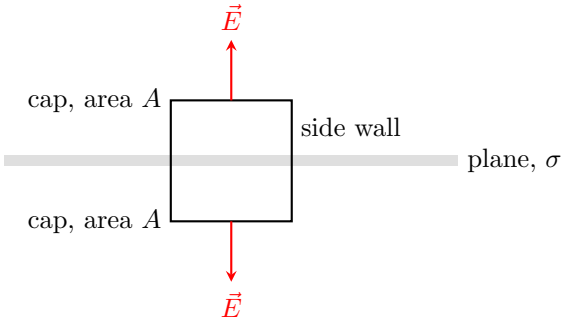


Figure 3: Side view of the pillbox straddling the plane.

$$2EA = \frac{\sigma A}{\epsilon_0},$$

so

$$E = \frac{\sigma}{2\epsilon_0}. \tag{13}$$

The field does not depend on the distance from the plane.

Example (Two planes)**.** *Superpose two sheets a distance d apart, each giving $\sigma/2\epsilon_0$ pointing away from itself.*

- *$+\sigma$ and $-\sigma$ (parallel plates): the fields add between the plates, $E = \sigma/\epsilon_0$ from $+$ to $-$, and cancel outside, $E = 0$.*
- *$+\sigma$ and $+\sigma$: the fields cancel between, $E = 0$, and add outside, $E = \sigma/\epsilon_0$ pointing away from both.*

Example (Infinite plane via Coulomb's law)**.** *The plane is still infinite, not a disk. We just describe it in polar coordinates (r', ϕ') centred directly below the field point at height z; letting r' run from 0 to ∞ covers the whole plane. Sideways components cancel in pairs, so each piece $dq = \sigma r'\, dr'\, d\phi'$ contributes only its $\hat{z}$ part, via $\cos\psi = z/\sqrt{r'^2 + z^2}$ with ψ the angle between $\hat{\mathfrak{r}}$ and $\hat{z}$:*

$$\vec{E} = \hat{z} \int_0^{2\pi} d\phi' \int_0^{\infty} \frac{\sigma}{4\pi\epsilon_0} \frac{r' z}{(r'^2 + z^2)^{3/2}}\, dr' = \frac{\sigma z}{2\epsilon_0} \left[-\frac{1}{\sqrt{r'^2 + z^2}} \right]_0^{\infty} \hat{z} = \frac{\sigma}{2\epsilon_0} \hat{z}.$$

Distance drops out because up close each piece pushes hard but mostly sideways, while far away each piece is weak but pushes more directly upward. The two effects cancel exactly.

3.2.4 Why the fall-off depends on the geometry

The fall-off is how fast E decreases with the distance r from the source. The examples above give

$$\text{point charge: } E \propto \frac{1}{r^2}, \qquad \text{infinite line: } E \propto \frac{1}{r}, \qquad \text{infinite plane: } E \propto r^0 \text{ (constant).}$$

The exponent counts how much charge a cone of field lines can "see" as it grows. Picture a narrow cone opening from the observation point toward the source: its cross section at distance r grows like r^2, which alone gives the $1/r^2$ dilution of a point charge. For extended sources, the charge inside the cone grows too:

- Line of charge: the cone intercepts a length $\propto r$, so $E \propto r/r^2 = 1/r$.
- Sheet of charge: the cone intercepts an area $\propto r^2$, so $E \propto r^2/r^2$, a constant. The infinite sheet has a uniform field.

Remark 18. *For two equal positive charges, no field lines cross the midplane: the field there is parallel to the plane and vanishes at the midpoint. Gauss' law still holds, but without symmetry it cannot solve for $\vec{E}$.*

3.3 The Curl of $\vec{E}$

Example (Line integral of a point charge)**.** *Nothing moves here. A point charge q sits fixed at the origin, and we compute the line integral of its field, $\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}$, along an arbitrary path from $\vec{a}$ to $\vec{b}$. (Times a test charge Q, this is the work the field would do on Q carried along that path.) In spherical coordinates $d\vec{l} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$, and since $\vec{E}$ is radial only the dr part survives:*

$$\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = \frac{q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_a} - \frac{1}{r_b} \right).$$

This depends only on the endpoints, so around a closed loop ($r_a = r_b$) it vanishes.

Theorem 9 (Electrostatic fields are curl-free)**.** *For any electrostatic field,*

$$\oint \vec{E} \cdot d\vec{l} = 0, \qquad \vec{\nabla} \times \vec{E} = 0. \tag{14}$$

Proof. Any static $\vec{E}$ is a sum of point-charge fields (superposition), so its loop integral is the sum of their loop integrals, each of which is zero. Stokes' theorem then gives $\iint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = 0$ for every surface, so $\vec{\nabla} \times \vec{E} = 0$. □

Remark 19. *Physically, $Q \oint \vec{E} \cdot d\vec{l}$ is the work done on a test charge around a closed loop. If it were nonzero, a charge could gain energy just by circling, violating energy conservation.*

Remark 20. *The proof only used that each point charge's field is central, $f(\mathfrak{r})\hat{\mathfrak{r}}$. So any central force law, e.g. a modified Coulomb's law with an extra factor $(1+\mathfrak{r}/\lambda)e^{-\mathfrak{r}/\lambda}$, still has $\vec{\nabla}\times\vec{E}=0$ and a potential.*

Remark 21 (Is this field electrostatic?)**.** *Check the curl. A field with parallel arrows whose length changes along the arrows, $\vec{E}=E(y)\hat{y}$, is curl-free, so it is allowed (with charge $\rho=\epsilon_0 E'(y)$). A shear field, $\vec{E}=E(y)\hat{x}$ with arrows reversing across a line, has $\vec{\nabla}\times\vec{E}=-E'(y)\hat{z}\neq 0$, so it is impossible in electrostatics.*

3.4 Electric Potential

Definition 25 (Electric potential)**.** *Since $\vec{\nabla}\times\vec{E}=0$, the irrotational fields theorem guarantees a scalar V with*

$$\vec{E}=-\vec{\nabla}V. \tag{15}$$

The minus sign is convention: positive charges are pushed toward lower potential energy.

Remark 22. *$\oint \vec{E}\cdot d\vec{l}=0$ holds only in electrostatics. A changing magnetic field creates an electric field that circulates in loops (Faraday's law, $\vec{\nabla}\times\vec{E}=-\partial\vec{B}/\partial t$), so a charge can gain net energy going around. This is how a generator drives current around a closed circuit. Energy is still conserved: it comes from whatever is changing $\vec{B}$.*

Proposition 9 (Potential from the field)**.**

$$V(\vec{r})=-\int_{ref}^{\vec{r}}\vec{E}\cdot d\vec{l}. \tag{16}$$

The path does not matter since the integral is path-independent[3]*. By convention the reference is at infinity, unless the charge distribution extends to infinity.*

Example (Point charge)**.** *Along a radial path,*

$$V(r)=-\frac{q}{4\pi\epsilon_0}\int_{\infty}^{r}\frac{dr'}{r'^2}=\frac{q}{4\pi\epsilon_0 r}.$$

Example (Uniform solid sphere)**.** *Using the field of Example 3.2.1, outside $V=\frac{q}{4\pi\epsilon_0 r}$. Inside, integrate in from infinity through both regions:*

$$V(r)=-\int_{\infty}^{R}\frac{q\,dr'}{4\pi\epsilon_0 r'^2}-\int_{R}^{r}\frac{q\,r'\,dr'}{4\pi\epsilon_0 R^3}=\frac{q}{8\pi\epsilon_0 R}\left(3-\frac{r^2}{R^2}\right).$$

[3] Changing the reference point adds a constant to V, which leaves $\vec{E}=-\vec{\nabla}V$ unchanged (gauge freedom). Statement (ii) of the irrotational fields theorem.

Proposition 10 (Potential of a distribution)**.** *By superposition,*

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\vec{r}')}{\mathfrak{r}}\, d\tau', \tag{17}$$

where $\rho(\vec{r}')d\tau' = dq$. Unlike the integral for $\vec{E}$, there is no $\hat{\mathfrak{r}}$, so computing V first and then $\vec{E} = -\vec{\nabla}V$ is often easier.

Problem 2 (Midterm drills)**.** *Find V (i) at the centre of a ring of radius a and charge Q; (ii) at $x = 2a$ for a line of charge Q from $x = -a$ to $x = a$.*

Solution 2. *(i) Every piece of the ring is a distance a away, so $V = \frac{Q}{4\pi\epsilon_0 a}$. Note $\vec{E} = 0$ there by symmetry, but $V \neq 0$.*

(ii) With $\lambda = Q/2a$ and $\mathfrak{r} = 2a - x'$,

$$V = \frac{\lambda}{4\pi\epsilon_0}\int_{-a}^{a} \frac{dx'}{2a - x'} = \frac{\lambda}{4\pi\epsilon_0}\ln\frac{3a}{a} = \frac{Q\ln 3}{8\pi\epsilon_0 a}.$$

Law 3 (Poisson's equation)**.** *Combining $\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$ with $\vec{E} = -\vec{\nabla}V$,*

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}. \tag{18}$$

3.5 Boundary Conditions

A surface charge σ makes $\vec{E}$ jump as you cross it. Gauss' law fixes the jump in the perpendicular part, and $\oint \vec{E}\cdot d\vec{l} = 0$ fixes the parallel part.

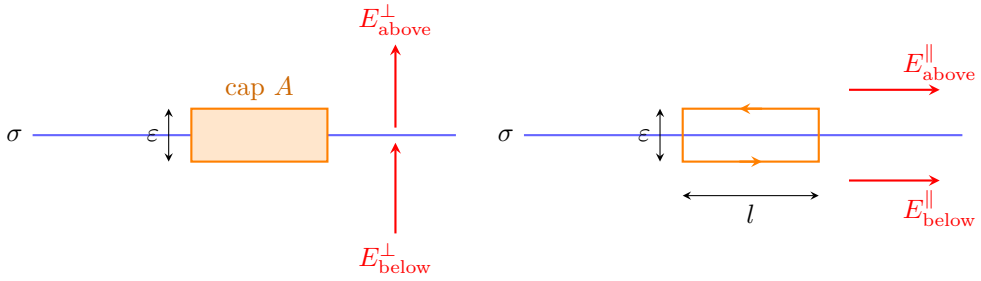


Figure 4: Side views of a charged surface. Left: a pillbox of cap area A and thickness ε gives the normal condition. Right: a thin rectangular loop of length l gives the tangential condition.

Proposition 11 (Normal component)**.** *Take a pillbox of cap area A straddling the surface and shrink its thickness $\varepsilon \to 0$, so the side wall carries no flux. Only the caps count, and the charge inside is σA:*

$$E^{\perp}_{above}A - E^{\perp}_{below}A = \frac{\sigma A}{\epsilon_0} \quad\Longrightarrow\quad E^{\perp}_{above} - E^{\perp}_{below} = \frac{\sigma}{\epsilon_0}.$$

Proposition 12 (Tangential component)**.** *Take a thin rectangular loop of length l straddling the surface, again with $\varepsilon \to 0$ so the short sides contribute nothing. Since $\oint \vec{E} \cdot d\vec{l} = 0$,*

$$E^{\parallel}_{above}\, l - E^{\parallel}_{below}\, l = 0 \quad \Longrightarrow \quad \vec{E}^{\parallel}_{above} = \vec{E}^{\parallel}_{below}.$$

Theorem 10 (Boundary condition on $\vec{E}$)**.** *With $\hat{n}$ the unit normal pointing from below to above, the two results combine into*

$$\vec{E}_{above} - \vec{E}_{below} = \frac{\sigma}{\epsilon_0}\hat{n}. \tag{19}$$

The normal component jumps by σ/ϵ_0; the tangential component is continuous.

Remark 23. *Check with the infinite plane: $\vec{E} = +\frac{\sigma}{2\epsilon_0}\hat{n}$ above and $-\frac{\sigma}{2\epsilon_0}\hat{n}$ below, a jump of exactly σ/ϵ_0. Conversely, $\vec{E} = +2\hat{z}$ above and $-2\hat{z}$ below the plane $z = 0$ means $\sigma = 4\epsilon_0$.*

Remark 24. *So $\vec{E}$ is not always continuous: it jumps wherever there is a surface charge. The condition holds on any surface, insulator or conductor.*

Theorem 11 (Boundary conditions on V)**.** *The potential is continuous across the surface,*

$$V_{above} = V_{below}, \tag{20}$$

because $V_{above} - V_{below} = -\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}$ along a path straight through the surface, and as the path shrinks to zero length the integral vanishes ($\vec{E}$ stays finite). Its slope is not continuous: substituting $\vec{E} = -\vec{\nabla}V$ into Theorem 10 gives $\vec{\nabla}V_{above} - \vec{\nabla}V_{below} = -\frac{\sigma}{\epsilon_0}\hat{n}$, or with the normal derivative $\frac{\partial V}{\partial n} \equiv \vec{\nabla}V \cdot \hat{n}$,

$$\frac{\partial V_{above}}{\partial n} - \frac{\partial V_{below}}{\partial n} = -\frac{\sigma}{\epsilon_0}. \tag{21}$$

Remark 25. *So V has a kink at a charged surface, like $|x|$ at $x = 0$: continuous, but its slope changes abruptly. $\vec{E}$, being the slope, jumps.*

3.6 Work and Energy in Electrostatics

Proposition 13 (Potential of a test charge)**.** *Bringing Q in from infinity against the field takes*

$$W = -Q\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{l} = QV(\vec{r}). \tag{22}$$

So the electric potential V is the potential energy per unit charge, in $\mathrm{J/C} = \mathrm{V}$.

Proposition 14 (Energy of N point charges)**.** *The first charge costs no work, since no field exists yet, and each later one feels the potential $V(\vec{r}_i)$ of those already placed, giving one term per pair:*

$$U = \frac{1}{2}\sum_{i\neq j}\frac{q_i q_j}{4\pi\epsilon_0|\vec{r}_i - \vec{r}_j|} = \frac{1}{2}\sum_i q_i V(\vec{r}_i), \tag{23}$$

where the $\frac{1}{2}$ corrects for counting each pair twice.

Remark 26. *U does not obey superposition, because it is quadratic in the charges rather than linear: doubling every charge doubles $\vec{E}$ but quadruples U. If two configurations with energies U_1 and U_2 are combined, the pair sum picks up new pairs (one charge from each), so $U = U_1 + U_2 + U_{int}$. In field language (Theorem 12), $E^2 = E_1^2 + E_2^2 + 2\vec{E}_1\cdot\vec{E}_2$, and the cross term is the interaction energy.*

Theorem 12 (Energy stored in the field)**.** *For a continuous distribution,*

$$U = \frac{1}{2}\int \rho V\, d\tau = \frac{\epsilon_0}{2}\int_{all\ space} E^2\, d\tau. \tag{24}$$

Definition 26 (Energy Density)**.** *From Theorem 12, the energy per unit volume stored in the field is*

$$u = \frac{\epsilon_0}{2}E^2. \tag{25}$$

Proof. Using the energy of N point charges, $U = \frac{1}{2}\sum_i q_i V(\vec{r}_i)$. For a continuous distribution this becomes $U = \frac{1}{2}\int \rho V\, d\tau$. Substituting $\rho = -\epsilon_0\nabla^2 V$ from Poisson's equation,

$$U = -\frac{\epsilon_0}{2}\int V\nabla^2 V\, d\tau.$$

To rewrite $V\nabla^2 V$, use the product rule $\partial_i(fg) = (\partial_i f)g + f(\partial_i g)$ with $f = V$ and $g = \partial_i V$, summed over i:

$$\underbrace{\sum_i \partial_i(V\partial_i V)}_{\vec{\nabla}\cdot(V\vec{\nabla}V)} = \underbrace{\sum_i(\partial_i V)^2}_{|\vec{\nabla}V|^2} + \underbrace{V\sum_i \partial_i\partial_i V}_{V\nabla^2 V} \quad\Longrightarrow\quad V\nabla^2 V = \vec{\nabla}\cdot(V\vec{\nabla}V) - |\vec{\nabla}V|^2.$$

Substituting, the overall minus sign flips both terms:

$$U = -\frac{\epsilon_0}{2}\int\left[\vec{\nabla}\cdot(V\vec{\nabla}V) - |\vec{\nabla}V|^2\right]d\tau = \frac{\epsilon_0}{2}\int|\vec{\nabla}V|^2\, d\tau - \frac{\epsilon_0}{2}\int\vec{\nabla}\cdot(V\vec{\nabla}V)\, d\tau.$$

The second integral vanishes. To see this, integrate over a huge sphere of radius R instead of all space, and let $R\to\infty$ at the end. The divergence theorem turns the integral into a flux through the sphere's surface, $\oint V\vec{\nabla}V\cdot d\vec{a}$. If the charge sits in a finite region, then far away $V\sim 1/R$ and $|\vec{\nabla}V|\sim 1/R^2$, so the integrand is $\sim 1/R^3$, while the area is $4\pi R^2$. The flux is therefore $\sim 1/R$, which goes to zero. With $\vec{E} = -\vec{\nabla}V$, $U = \frac{\epsilon_0}{2}\int E^2\, d\tau$. □

Remark 27 (Self-energy)**.** *The two formulas do not always agree. The pair sum only counts the work of bringing different charges together ($i \neq j$); it ignores the work needed to build each point charge itself. The field integral counts everything, including that self-energy. So $\frac{\epsilon_0}{2}\int E^2\,d\tau$ is always positive (since $E^2 \geq 0$), while the pair sum can be negative, e.g. $+q$ and $-q$ at distance d give $U = -\frac{q^2}{4\pi\epsilon_0 d}$. For a true point charge the self-energy is infinite: near it $E^2 \sim 1/r^4$, and $\int \frac{1}{r^4}\,r^2\,dr$ blows up at $r = 0$. So use the pair sum for point charges and the field integral for continuous distributions.*

Problem 3 (Midterm drill)**.** *Two plates of area A, a distance d apart, carry $+\sigma$ and $-\sigma$ ($A \gg d^2$). Find the energy.*

Solution 3. *The field is σ/ϵ_0 between the plates and zero outside, so*

$$U = \frac{\epsilon_0}{2}\left(\frac{\sigma}{\epsilon_0}\right)^2 Ad = \frac{\sigma^2 Ad}{2\epsilon_0}.$$

Problem 4 (Problem Set 4, Q4)**.** *Concentric shells of radii $a < b$ carry $+q$ and $-q$. Find the energy.*

Solution 4. *By Gauss, $\vec{E} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$ only for $a < r < b$, and zero elsewhere. With $d\tau = 4\pi r^2 dr$,*

$$U = \frac{\epsilon_0}{2}\int_a^b \left(\frac{q}{4\pi\epsilon_0 r^2}\right)^2 4\pi r^2\,dr = \frac{q^2}{8\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right).$$

3.7 Conductors

Definition 27 (Conductor)**.** *A conductor is a material with charges that are free to move (in a metal, the electrons). Placed in an external field, the free charges drift: negative ones pile up on one side and positive ones on the other, until the field they create, $\vec{E}_{int}$, exactly cancels $\vec{E}_{ext}$ inside. This happens almost instantly, and the result is electrostatic equilibrium.*

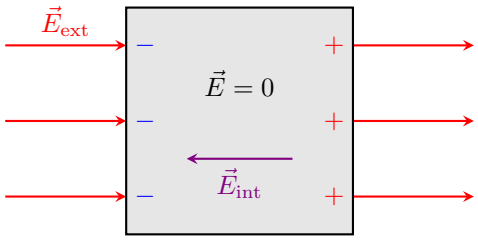


Figure 5: A conductor in an external field. The induced charges create $\vec{E}_{\text{int}}$, which cancels $\vec{E}_{\text{ext}}$ inside the metal. Outside, field lines end on the $-$ charges and start again from the $+$ charges.

Proposition 15 (Properties of a conductor in equilibrium)**.** *(i) $\vec{E} = 0$ inside, otherwise the free charges would still be moving.*

(ii) $\rho = 0$ inside, since $\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E} = 0$.

(iii) Any net charge therefore sits on the surface.

(iv) The conductor is an equipotential: for any two points in it, $V(\vec{b}) - V(\vec{a}) = -\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = 0$.

(v) Just outside, $\vec{E}$ is perpendicular to the surface. A tangential part would push the surface charges along the surface, so they would not be in equilibrium.

Remark 28. *Grounding a conductor connects it to $V(\infty) = 0$, so its whole body, not just its surface, sits at $V = 0$. Charge flows to or from the ground as needed to make this happen.*

Proposition 16 (Field at the surface of a conductor)**.** *Theorem 10 with $\vec{E}_{below} = 0$ (inside the metal) gives, just outside,*

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}, \tag{26}$$

or in terms of the potential, $\frac{\partial V}{\partial n} = -\frac{\sigma}{\epsilon_0}$. Where σ is large, the field outside is strong.

Remark 29. *Compare the infinite sheet, $E = \sigma/2\epsilon_0$. A sheet sends half its flux up and half down; a conductor has no field inside, so all the flux goes out one side and the field is doubled.*

Example (Neutral conductor near a charged rod)**.** *A neutral metal block sits near an insulating rod with uniform positive charge. Negative charge is induced on the side facing the rod and an equal positive charge on the far side. The negative charges are closer, and Coulomb's force falls off with distance, so the attraction beats the repulsion: the neutral conductor is pulled toward the rod.*

Example (Conducting versus insulating disk)**.** *For a disk with uniform σ (an insulator), the disk is not an equipotential and the field lines are not perpendicular to it. That arrangement cannot last on a conductor: the field along the surface pushes charge outward, i.e. "downhill" in potential, and it piles up at the rim. So a charged conducting disk has a higher σ near its edge than near its centre, its surface is an equipotential, and its field lines leave it at right angles.*

Remark 30. *In general, charge on a conductor concentrates where the surface is most sharply curved (edges, corners, points). Since $E = \sigma/\epsilon_0$ there, the field is strongest at sharp points, which is how lightning rods work.*

Remark 31 (Cavities)**.** *If a conductor has an empty cavity, the field inside the cavity is zero: its wall is an equipotential, and a solution of Laplace's equation with constant boundary values is constant (no maxima or minima, see the Laplace's equation section). This is a Faraday cage. If a charge q sits in the cavity, it induces $-q$ on the cavity wall and $+q$ on the outer surface (for a neutral conductor), and the outside cannot tell how the charge inside is arranged. In particular, for a sphere the outer charge spreads uniformly even if the cavity is off centre, so the field outside is that of a point charge at the centre of the sphere, not of the cavity.*

Example (Concentric conducting shells)**.** *A metal sphere of radius R with charge q sits inside a thick neutral metal shell of inner radius a and outer radius b.*

(i) A Gaussian sphere inside the shell's metal ($a < r < b$) has $\vec{E} = 0$ on it, so it encloses zero charge: the inner surface carries $-q$. The shell is neutral, so the outer surface carries $+q$. By symmetry each is uniform:

$$\sigma_R = \frac{q}{4\pi R^2}, \qquad \sigma_a = -\frac{q}{4\pi a^2}, \qquad \sigma_b = \frac{q}{4\pi b^2}.$$

(ii) The field is $\frac{q}{4\pi\epsilon_0 r^2}\hat{r}$ for $r > b$ and $R < r < a$, and zero in both metals. Integrating in from infinity,

$$V(0) = -\int_\infty^b \frac{q\,dr}{4\pi\epsilon_0 r^2} - \int_a^R \frac{q\,dr}{4\pi\epsilon_0 r^2} = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{b} + \frac{1}{R} - \frac{1}{a}\right).$$

(iii) Grounding the outer surface sets $V(b) = 0 = V(\infty)$, so there can be no field outside and $\sigma_b = 0$. The Gauss argument in (i) is unchanged, so σ_R and σ_a stay the same, and the first integral in (ii) disappears: $V(0) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{R} - \frac{1}{a}\right)$.

(iv) The outer surface now has zero charge; the $+q$ flowed away to the ground, leaving the shell with net charge $-q$. If the outer surface kept a charge Q', the total charge inside a sphere just outside b would be Q', giving $V(b) = \frac{Q'}{4\pi\epsilon_0 b} \neq 0$.

Problem 5 (Problem Set 4, Q3)**.** *A neutral conducting sphere of radius R has two spherical cavities, of radii a and b, with point charges q_a and q_b at their centres. (i) Find σ_a, σ_b, σ_R; (ii) the field outside; (iii) the field in each cavity; (iv) the force on q_a and q_b; (v) what changes if a charge q_c is brought near the outside?*

Solution 5. *(i) Each cavity wall carries minus the charge inside it, spread uniformly since the charge is at the centre, and the outer surface carries the rest. The metal shields the outer surface from the cavities, so its charge spreads uniformly as on an isolated sphere:*

$$\sigma_a = -\frac{q_a}{4\pi a^2}, \qquad \sigma_b = -\frac{q_b}{4\pi b^2}, \qquad \sigma_R = \frac{q_a + q_b}{4\pi R^2}.$$

(ii) *Outside,* $\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q_a+q_b}{r^2}\hat{r}$*, with* r *measured from the centre of the big sphere.*

(iii) *Inside cavity* a*,* $\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q_a}{r_a^2}\hat{r}_a$*, with* $\vec{r}_a$ *measured from the centre of the cavity; the same holds for cavity* b*.*

(iv) *The force on* q_a *and* q_b *is zero. Apart from its own field,* q_a *only feels the uniform shell of charge on its cavity wall, which produces no field inside it; everything beyond the wall is shielded by the metal.*

(v) *Bringing* q_c *near the outside changes* σ_R *(it is no longer uniform) and the field outside, but nothing inside:* σ_a*,* σ_b*, the cavity fields and the forces on* q_a *and* q_b *stay the same. The conductor shields in both directions.*

Problem 6 (Midterm 2025, Q3)**.** *A thick conducting shell (inner radius* a*, outer radius* b*, empty cavity) has net charge* $-Q$*. (a) Where is the charge? (b) Sketch* $E_r(r)$*. (c) Sketch* $V(r)$ *with* $V(\infty) = 0$*. (d) A charge* $+q$ *with* $q > Q$ *is placed outside: sketch the negative charge. (e) Find the total charge on the inner surface and (f) on the outer surface.*

Solution 6. *(a)* $\rho = 0$ *in the metal. A Gaussian sphere in the metal has* $\vec{E} = 0$*, and the cavity is empty, so the inner surface has* $\sigma_a = 0$*. All the charge is on the outer surface:* $\sigma_b = -\frac{Q}{4\pi b^2}$*.*

(b) $E_r = 0$ *for* $r < b$ *(cavity and metal);* $E_r = -\frac{Q}{4\pi\epsilon_0 r^2}$ *for* $r > b$ *(pointing inward). It jumps at* b *to its most negative value* $-\frac{Q}{4\pi\epsilon_0 b^2}$*, then rises toward* 0 *as* $1/r^2$*.*

(c) $V = -\frac{Q}{4\pi\epsilon_0 r}$ *for* $r > b$*, and constant* $V = -\frac{Q}{4\pi\epsilon_0 b}$ *for all* $r \leq b$*. It is continuous, with a kink at* b*.*

(d) *Negative charge crowds onto the part of the outer surface facing* $+q$*. Since* $q > Q$*, the side facing away ends up positive.*

(e) $q_{tot,a} = 0$*. A Gaussian surface inside the metal still has* $\vec{E} = 0$ *on it and encloses only the empty cavity, so the inner wall carries no net charge (and in fact none at all, since the cavity field is zero).*

(f) $q_{tot,b} = -Q$ *by charge conservation; only its distribution changed.*

4 Potentials

September 18-25, 2026

4.1 Laplace's Equation

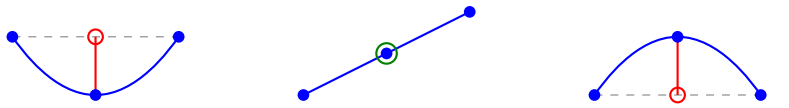
We want V because $\vec{E} = -\vec{\nabla}V$. The integral $V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho}{\mathfrak{r}}\, d\tau'$ is exact but often impossible to do, and the multipole expansion is only an approximation. Often we do not even know ρ everywhere: we know the potentials of some conductors instead, and the charge on them is whatever it has to be. In the empty space between the charges, Poisson's equation simplifies.

Law 4 (Laplace's equation)**.** *In a region with no charge,*

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0. \tag{27}$$

Remark 32. *Point charges, charged objects and conductors are where* $\nabla^2 V = -\rho/\epsilon_0$ *holds with* $\rho \neq 0$*. Everywhere else* $\nabla^2 V = 0$*, and the charges only enter as boundary conditions on that empty region.*

Remark 33 (The Laplacian compares a point with its surroundings)**.** *In one dimension,* $V'' = 0$ *gives a straight line,* $V = mx + b$*, and then* $V(x) = \frac{1}{2}[V(x+a) + V(x-a)]$*: every point is the average of its neighbours. In general* $\nabla^2 V$ *measures how far a point sits from the average of its surroundings, and since* $\nabla^2 V = -\rho/\epsilon_0$*, that tells you the sign of the charge there.*



below the average
$\nabla^2 V > 0$, so $\rho < 0$

equal to the average
$\nabla^2 V = 0$, no charge

above the average
$\nabla^2 V < 0$, so $\rho > 0$

Figure 6: The middle point compared with the average of its two neighbours (open circle).

Theorem 13 (Mean value property)**.** *If* $\nabla^2 V = 0$*, the value of* V *at any point equals its average over any sphere of radius* R *centred on that point (containing no charge):*

$$V(\vec{r}_0) = \frac{1}{4\pi R^2} \oint_{sphere} V \, da. \tag{28}$$

Corollary 13.1 (No local maxima or minima)**.** *A solution of Laplace's equation has no local maxima or minima inside the region; extreme values occur only on the boundary. A maximum would be larger than every point around it, so it could not equal their average.*

Theorem 14 (Earnshaw's theorem)**.** *A charged particle cannot be held in stable equilibrium by electrostatic forces alone.*

Proof. Stable equilibrium for a charge q needs its potential energy qV to have a local minimum at that point. But the other charges' V obeys Laplace's equation there, so it has no local minimum. Concretely, if $\frac{\partial^2 V}{\partial x^2} > 0$ (stable along x), then $\frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} < 0$, so the charge escapes sideways. For example, a charge between two equal charges is stable along the line joining them but slips out perpendicular to it. □

Theorem 15 (Uniqueness theorem)**.** *The solution of Laplace's equation in a volume is uniquely determined if* V *is specified on the whole boundary of that*

volume. Similarly, if the region contains conductors, $\vec{E}$ is uniquely determined once the charge density ρ and the total charge on each conductor are given.

Remark 34. *The significance: any function that satisfies Laplace's equation and the boundary conditions is* the *answer, no matter how it was found. Guessing is allowed, which is what the method of images and separation of variables exploit.*

Proposition 17 (Method of relaxation)**.** *On a grid, the mean value property says each cell equals the average of its neighbours. In two dimensions,*

$$V_{i,j} = \frac{1}{4}\left(V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1}\right).$$

This gives a numerical method:

1. *Initialize the grid with a guess for V.*
2. *Sweep through the grid, replacing each value by the average of its nearest neighbours, keeping the boundary values fixed.*
3. *Repeat until the values stop changing.*

4.2 The Method of Images

Remark 35 (A solution in search of a problem)**.** *Maxwell pointed out that the inverse problem is easier: given a potential, finding conductors that fit it is simpler than finding the potential for given conductors. Any equipotential surface of a known potential can be replaced by a conductor held at that potential, and by uniqueness the field outside it does not change.*

Example (Point charge above a grounded plane)**.** *A charge q sits at height d above an infinite grounded conducting plane at $z = 0$. For $z > 0$ we need $\nabla^2 V = 0$ (apart from at q), $V = 0$ on the plane, and $V \to 0$ far away. Forget the plane and put an image charge $-q$ at $z = -d$ instead. Every point of the plane is equidistant from the two charges, so this pair gives $V = 0$ there, and it has the same charge in the region $z > 0$. By uniqueness it is the answer for $z \geq 0$:*

$$V(x,y,z) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{\sqrt{x^2+y^2+(z-d)^2}} - \frac{q}{\sqrt{x^2+y^2+(z+d)^2}}\right].$$

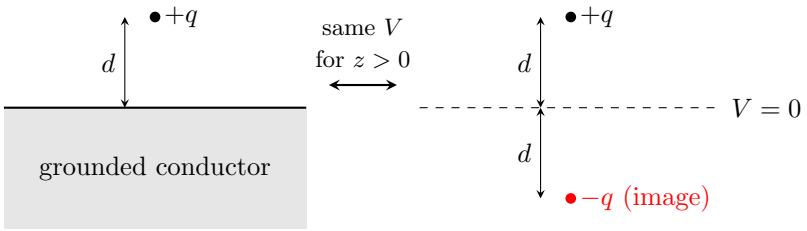


Figure 7: The real problem (left) and the image problem (right) have the same charge and the same boundary values in $z > 0$, so they have the same potential there.

Remark 36. *The image charge is not real; it is bookkeeping. Physically, the field comes from the negative surface charge induced on the plane, and the image reproduces its effect in the region we care about. The formula says nothing about $z < 0$, where $V = 0$ inside the conductor. The image must sit outside the region being solved, otherwise it would change ρ there.*

Remark 37. *The induced charge on the plane totals $-q$, and q is attracted to the plane with the force of its image, but the energy is only half of the image pair's, since there is no field in $z < 0$.*

Example (Two charges above a grounded plane)**.** *Charges $+Q$ at height d and $+3Q$ at height $2d$ sit above a grounded plane. Each charge on its own is handled by one image, so by superposition we need two: $-Q$ at $-d$ and $-3Q$ at $-2d$. The method works whenever the problem is linear (superposition) and the boundary values come out right (uniqueness).*

4.3 Separation of Variables: Cartesian

Proposition 18 (Separation)**.** *Look for solutions $V(x, y, z) = X(x)Y(y)Z(z)$. Substituting into $\nabla^2 V = 0$ and dividing by V,*

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = 0.$$

Each term depends on a different variable, which can be varied alone, so each must be a constant: $X'' = c_x X$, $Y'' = c_y Y$, $Z'' = c_z Z$ with $c_x + c_y + c_z = 0$. One PDE becomes three ODEs.

Remark 38. *Most solutions are not separable (e.g. $1/r$), but Laplace's equation is linear, so sums of separable solutions can build general ones.*

Proposition 19 (Three kinds of solution)**.** *For each direction,*

$$c = +k^2 : \ X = Ae^{kx} + Be^{-kx}, \qquad c = -k^2 : \ X = A\sin kx + B\cos kx, \qquad c = 0 : \ X = A + Bx.$$

Since the constants sum to zero, they cannot all have the same sign unless all vanish.

Remark 39 (Which direction gets which sign)**.** *A sinusoid crosses zero infinitely often; an exponential (or* cosh, sinh*) at most once. So a direction in which V must vanish at two places gets the sinusoid, and the other gets the exponential. Symmetry picks the parity: even* $\Rightarrow$ cos, cosh*; odd* $\Rightarrow$ sin, sinh. cosh *never vanishes;* sinh *vanishes only at* 0*.*

Example (Square pipe)**.** *A long pipe along z with square cross section: sides $x = \pm a/2$ grounded, top and bottom $y = \pm a/2$ held at V_0. Nothing depends on z, so $c_z = 0$ and $c_x = -c_y$. V vanishes at two values of x, so X is sinusoidal and Y hyperbolic; the problem is even in both x and y, so*

$$V = C \cos kx \, \cosh ky.$$

$V(\pm a/2, y) = 0$ quantises k: $\cos(ka/2) = 0 \Rightarrow k = (2n+1)\pi/a$. Superposing,

$$V(x,y) = \sum_{n=0}^{\infty} C_n \cos \frac{(2n+1)\pi x}{a} \cosh \frac{(2n+1)\pi y}{a}.$$

Setting $y = a/2$, multiplying by $\cos \frac{(2m+1)\pi x}{a}$ and integrating over x (Fourier's trick, using orthogonality $\int_{-a/2}^{a/2} \cos\cos \, dx = \frac{a}{2}\delta_{mn}$) picks out one term:

$$V(x,y) = \frac{4V_0}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos \frac{(2n+1)\pi x}{a} \, \frac{\cosh\left[(2n+1)\pi y/a\right]}{\cosh\left[(2n+1)\pi/2\right]}.$$

Check: rotating by 90° swaps hot and grounded faces, so $V(0,0) = V_0/2$ exactly; the series gives $0.5074\,V_0$ with one term. The centre is a saddle, as $\nabla^2 V = 0$ requires. Near the corners the series overshoots (Gibbs phenomenon), because the boundary data jumps there.

Remark 40 (Summary)**.**

1. *Separate: one PDE becomes ODEs with constants summing to zero.*
2. *Choose signs: two or more zeros in a direction $\Rightarrow$ sinusoid; otherwise exponential/hyperbolic.*
3. *Use symmetry to keep only even or odd functions.*
4. *Impose the $V = 0$ boundaries: these quantise k.*
5. *Superpose and use Fourier's trick for the coefficients.*
6. *Uniqueness guarantees the result is the answer.*

4.4 Separation of Variables: Spherical

Proposition 20 (Axially symmetric solutions)**.** *If nothing depends on ϕ, separating $V(r,\theta) = R(r)\Theta(\theta)$ in the spherical Laplacian gives radial solutions r^l and $r^{-(l+1)}$ and angular solutions the Legendre polynomials $P_l(\cos\theta)$, so*

$$V(r,\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos\theta), \tag{29}$$

with $P_0 = 1$, $P_1 = x$, $P_2 = (3x^2 - 1)/2$, *and orthogonality* $\int_0^\pi P_m P_n \sin\theta \, d\theta = \frac{2}{2m+1}\delta_{mn}$.

Proposition 21 (Sphere with given surface potential $V_0(\theta)$)**.**

- *Inside,* V *must be finite at* $r = 0$*, so* $B_l = 0$*.*
- *Outside,* $V \to 0$ *at infinity, so* $A_l = 0$*.*
- *Write* $V_0(\theta) = \sum_l C_l P_l(\cos\theta)$ *(by inspection if it is a simple polynomial in* $\cos\theta$*, else by orthogonality). Continuity at* R *then gives*

$$V_{in} = \sum_l C_l \left(\frac{r}{R}\right)^l P_l(\cos\theta), \qquad V_{out} = \sum_l C_l \left(\frac{R}{r}\right)^{l+1} P_l(\cos\theta).$$

- *The surface charge comes from the jump in slope,* $\sigma = -\epsilon_0 \left(\frac{\partial V_{out}}{\partial r} - \frac{\partial V_{in}}{\partial r}\right)_{r=R}$*, giving*

$$\sigma(\theta) = \frac{\epsilon_0}{R} \sum_l (2l+1) C_l P_l(\cos\theta).$$

Remark 41. *Rewrite powers of* $\cos\theta$ *in Legendre polynomials, e.g.* $\cos^2\theta = \frac{2}{3}P_2 + \frac{1}{3}P_0$*. Each* l *outside is a multipole:* $l = 0$ *gives* $V \propto 1/r$ *with total charge* $Q = 4\pi\epsilon_0 R C_0$*.*

Problem 7 (Midterm 2025, Q5)**.** *A spherical shell of radius* R *has* $V(R, \theta) = V_0(1 + \cos\theta)$*, no charge elsewhere, and* $V(\infty) = 0$*. Find* V *outside,* V *inside, and* $\sigma(\theta)$*.*

Solution 7. $V_0(1 + \cos\theta) = V_0 P_0 + V_0 P_1$*, so* $C_0 = C_1 = V_0$ *and all others vanish:*

$$V_{out} = V_0 \left(\frac{R}{r} + \frac{R^2}{r^2}\cos\theta\right), \qquad V_{in} = V_0 \left(1 + \frac{r}{R}\cos\theta\right),$$

$$\sigma(\theta) = \frac{\epsilon_0 V_0}{R}\left(1 + 3\cos\theta\right).$$

Check: $V_{in} = V_0 + V_0 z/R$ *is a uniform field inside, and* V_{out} *is a monopole plus a dipole.*

4.5 Multipole Expansion

Example (Two charges on the z axis)**.** *Place* q_1 *at* $z = d/2$ *and* q_2 *at* $z = -d/2$*, and measure* θ *from the* z *axis.*

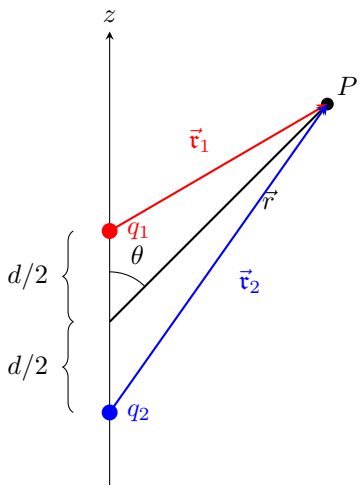


Figure 8: Two charges on the z axis and a far field point P at distance r and angle θ.

Then

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{\mathfrak{r}_1} + \frac{q_2}{\mathfrak{r}_2}\right).$$

To get $\mathfrak{r}_1$, write $\vec{\mathfrak{r}}_1 = \vec{r} - \frac{d}{2}\hat{z}$ and square it, using $\vec{r}\cdot\hat{z} = r\cos\theta$ (this is just the law of cosines):

$$\mathfrak{r}_1^2 = r^2 + \left(\frac{d}{2}\right)^2 - 2r\frac{d}{2}\cos\theta = r^2 + \left(\frac{d}{2}\right)^2 - rd\cos\theta.$$

For q_2 the source sits at $-\frac{d}{2}\hat{z}$, so the cross term flips sign. Both cases in one line, top sign for q_1 and bottom sign for q_2:

$$\mathfrak{r}_{1,2}^2 = r^2 + \left(\frac{d}{2}\right)^2 \mp rd\cos\theta.$$

Writing $1/\mathfrak{r}_{1,2} = \frac{1}{r}(1+\epsilon_\mp)^{-1/2}$ with $\epsilon_\mp = \frac{d}{2r}\left(\frac{d}{2r} \mp 2\cos\theta\right)$ and expanding for $d \ll r$, $(1+\epsilon)^{-1/2} = 1 - \frac{1}{2}\epsilon + \frac{3}{8}\epsilon^2 - \ldots$, gives

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\left[\underbrace{\frac{q_1+q_2}{r}}_{monopole} + \underbrace{\frac{(q_1-q_2)d\cos\theta}{2r^2}}_{dipole} + \underbrace{\frac{(q_1+q_2)d^2}{4}\frac{3\cos^2\theta - 1}{2r^3}}_{quadrupole} + \ldots\right].$$

Each term gains a power of $1/r$, so $E \sim 1/r^2,\ 1/r^3,\ 1/r^4, \ldots$

Remark 42. *Setting $q_1 = -q_2$ kills the monopole term, leaving $V \sim 1/r^2$ and $E \sim 1/r^3$ at leading order.*

Definition 28 (Dipole Moment)**.** *For charges $+q$ and $-q$ separated by $\vec{d}$ (pointing from $-q$ to $+q$), the dipole moment is*

$$\vec{p} = q\vec{d}, \qquad [\vec{p}] = \mathrm{C} \cdot \mathrm{m}. \tag{30}$$

For any distribution,

$$\vec{p} = \sum_i q_i \vec{r}_i = \int \vec{r}' \rho(\vec{r}')\, d\tau', \tag{31}$$

which reduces to $q\vec{d}$ for the pair. It measures how far apart the positive and negative charge sit, and sets the dipole term of the potential,

$$V_{dip} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}. \tag{32}$$

Proposition 22 (Dipole field)**.** *For $\vec{p} = p\hat{z}$, $V_{dip} = \frac{p\cos\theta}{4\pi\epsilon_0 r^2}$, and $\vec{E} = -\vec{\nabla}V$ in spherical coordinates gives*

$$\vec{E}_{dip}(r,\theta) = \frac{p}{4\pi\epsilon_0 r^3} \left(2\cos\theta\, \hat{r} + \sin\theta\, \hat{\theta} \right). \tag{33}$$

Definition 29 (Ideal dipole)**.** *Take $q_1 = q$ and $q_2 = -q$. The monopole and quadrupole terms vanish, and the dipole term is $\frac{1}{4\pi\epsilon_0} \frac{qd\cos\theta}{r^2}$. Every term in the expansion carries a factor $q\, d^{\,n}$, which we can write as $(qd)\, d^{\,n-1}$. Now shrink $d \to 0$ while raising $q \to \infty$ so that the product qd stays fixed:*

- *The dipole term ($n = 1$) contains only qd, so it is unchanged.*
- *Every later term carries at least one extra factor of d, so it goes to zero.*

(q must grow, otherwise the dipole term would vanish too.) What is left is a point object with dipole moment $\vec{p} = q\vec{d}$ and exactly $V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p}\cdot\hat{r}}{r^2}$.

Remark 43. *The names count charges in powers of 2: monopole (1), dipole (2), quadrupole (4), octupole (8), and so on. Each is built from two opposite copies of the previous one, placed close together so the previous order cancels. For example, two opposite dipoles give a quadrupole with $V \sim 1/r^3$.*

Theorem 16 (Multipole expansion)**.** *For r large compared with the size of the distribution,*

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\vec{p}\cdot\hat{r}}{r^2} + \frac{\sum_{i,j} \hat{r}_i Q_{ij} \hat{r}_j}{r^3} + \dots \right), \tag{34}$$

with

$$Q = \int \rho(\vec{r}')\, d\tau', \qquad \vec{p} = \int \vec{r}' \rho(\vec{r}')\, d\tau', \qquad Q_{ij} = \int \frac{1}{2} \left(3r'_i r'_j - r'^2 \delta_{ij} \right) \rho(\vec{r}')\, d\tau'.$$

Griffiths writes the same series with Legendre polynomials; see Appendix A.1.

Remark 44. *Each term is smaller than the previous one by a factor of roughly d/r, where d is the size of the charge distribution. Far away ($r \gg d$) the first nonzero term dominates and the rest are small corrections. Close up ($r \sim d$) the terms no longer shrink and the expansion is useless. So to find the leading behaviour: compute Q; if it vanishes, compute $\vec{p}$; if that vanishes too, the leading term is the quadrupole.*

Remark 45 (Origin dependence)**.** *Shifting the origin by $\vec{a}$ changes $\vec{p} \to \vec{p} - Q\vec{a}$, so $\vec{p}$ is well defined only when $Q = 0$ (a neutral molecule, not an ion). In general only the first non-vanishing moment is origin-independent.*

Problem 8 (Midterm drill)**.** *Charges $3q$ at $a\hat{z}$, q at $-a\hat{z}$, and $-2q$ at $\pm a\hat{y}$. (i) Sketch how to find the energy. (ii) Find V to dipole order.*

Solution 8. *(i) Sum $\frac{q_i q_j}{4\pi\epsilon_0 \mathfrak{r}_{ij}}$ over the 6 pairs: the two z charges are $2a$ apart, as are the two y charges, and each z charge is $\sqrt{2}a$ from each y charge:*

$$U = \frac{1}{4\pi\epsilon_0 a}\left[\frac{3q^2}{2} + \frac{4q^2}{2} + \frac{2(3q)(-2q) + 2(q)(-2q)}{\sqrt{2}}\right] = \frac{q^2}{4\pi\epsilon_0 a}\left(\frac{7}{2} - 8\sqrt{2}\right).$$

(ii) $Q = 3q + q - 4q = 0$, and $\vec{p} = 3qa\hat{z} - qa\hat{z} - 2qa\hat{y} + 2qa\hat{y} = 2qa\hat{z}$, so

$$V \approx \frac{1}{4\pi\epsilon_0}\frac{2qa\cos\theta}{r^2}.$$

Problem 9 (Midterm 2025, Q4)**.** *Charges $+q$ and $-q$, a distance a apart, sit a height d above a conductor filling $z < 0$. (a) Find $\vec{E}$ and V for $z < 0$. (b) Describe how to find the exact V for $z > 0$. (c) Find how V and $|\vec{E}|$ fall off for $r \gg d$.*

Solution 9. *(a) $\vec{E} = 0$ inside the conductor, so V is constant, independent of z ($V = 0$, matching the plane at infinity).*

(b) Method of images: place $-q$ at depth d directly below $+q$, and $+q$ directly below $-q$. The four charges give $V = 0$ on $z = 0$, and the images lie outside $z > 0$, so by uniqueness

$$V = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{4}\frac{q_i}{|\vec{r} - \vec{r}_i|} \quad \text{for } z > 0.$$

(c) Total charge is 0. The real pair has $\vec{p} = qa\hat{x}$ and the image pair $-qa\hat{x}$, so the dipole moment also cancels. The leading term is the quadrupole: $V \sim 1/r^3$ and $|\vec{E}| \sim 1/r^4$.

5 Electric Fields in Matter

6 Magnetostatics

6.1 Current Density

Definition 30 (Current density)**.** *The current I through a wire is not a local quantity, since a wire has finite thickness. The local version is the current density*

$$\vec{J} = \frac{d\vec{I}}{da_\perp},$$

where $da_\perp$ is the differential cross section perpendicular to the flow.

Remark 46. *Applying the divergence theorem (Theorem 4) to $\vec{J}$ is how conservation of charge gets written down: the charge inside V can only decrease by flowing out through S,*

$$\frac{d}{dt}\iiint_V \rho\, dV = -\oiint_S \vec{J}\cdot d\vec{A} = -\iiint_V \vec{\nabla}\cdot\vec{J}\, dV \quad \Longrightarrow \quad \frac{\partial\rho}{\partial t} = -\vec{\nabla}\cdot\vec{J}.$$

7 Magnetic Fields in Matter

8 Electrodynamics

A Appendix

Remark 47 (Compactified dimensions)**.** *Suppose an extra dimension exists but is curled up, the way a flat sheet with coordinates (x, y) can be rolled into a cylinder whose second coordinate w is periodic with a small radius. At distances much smaller than that radius, field lines spread out into all the dimensions and the field falls off faster than $1/r^2$. At distances much larger than it, there is only room to spread out in the large dimensions and the usual $1/r^2$ is recovered. So a measured deviation from the inverse square law at short range would be evidence for extra dimensions.*

This has been tested for gravity by the Eöt-Wash torsion balance experiments (2004, 2007, 2020), which checked the $1/r^2$ law down to separations of $52\,\mu$m. Newtonian gravity fit the data, ruling out gravitational strength Yukawa interactions with ranges above $38.6\,\mu$m at 95% confidence.

A.1 Multipole Expansion with Legendre Polynomials

Griffiths' form of the multipole expansion, with α the angle between $\vec{r}$ and $\vec{r}'$ and P_n the Legendre polynomials:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0}\sum_{n=0}^{\infty}\frac{1}{r^{n+1}}\int (r')^n P_n(\cos\alpha)\rho(\vec{r}')\,d\tau'.$$

$P_0 = 1$ gives the monopole, $P_1 = \cos\alpha$ the dipole (since $r'\cos\alpha = \vec{r}'\cdot\hat{r}$), and $P_2 = \frac{1}{2}(3\cos^2\alpha - 1)$ the quadrupole.

A.2 Proving the Identities

Curl of a Curl. Set $\vec{G} \equiv \vec{\nabla}\times(\vec{\nabla}\times\vec{F})$. Then,

$$G_i = \sum_{j,k}\varepsilon_{ijk}\partial_j\sum_{m,n}\varepsilon_{kmn}\partial_m F_n = \sum_{j,k,m,n}\varepsilon_{ijk}\varepsilon_{kmn}\partial_j\partial_m F_n.$$

Using the cyclic permutation $\varepsilon_{ijk} = \varepsilon_{kij}$, so that the summed index k sits first in both epsilons, the epsilon-delta identity 5 gives

$$\sum_k \varepsilon_{kij}\varepsilon_{kmn} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}.$$

Reading off the proposition with the summed index k removed:

$$G_i = \sum_{j,m,n}(\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})\,\partial_j\partial_m F_n.$$

Each Kronecker delta collapses one sum by setting its indices equal. In the first term δ_{im} forces $m = i$ and δ_{jn} forces $n = j$; in the second, δ_{in} forces $n = i$ and δ_{jm} forces $m = j$:

$$G_i = \sum_j \partial_j\partial_i F_j - \sum_j \partial_j\partial_j F_i \overset{4}{=} \partial_i\sum_j\partial_j F_j - \sum_j\partial_j\partial_j F_i = \left(\vec{\nabla}(\vec{\nabla}\cdot\vec{F})\right)_i - (\nabla^2\vec{F})_i.$$

$$\therefore \vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}.$$

□

Curl of a Gradient. Let f be twice continuously differentiable. With $(\vec{\nabla} f)_k = \partial_k f$,

$$\left(\vec{\nabla} \times (\vec{\nabla} f)\right)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f.$$

The indices j and k are summed over, so they can be renamed $j \leftrightarrow k$:

$$\left(\vec{\nabla} \times (\vec{\nabla} f)\right)_i = \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f.$$

Now use antisymmetry, $\varepsilon_{ikj} = -\varepsilon_{ijk}$, and the fact that partial derivatives commute, $\partial_k \partial_j f = \partial_j \partial_k f$:

$$\left(\vec{\nabla} \times (\vec{\nabla} f)\right)_i = -\sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f = -\left(\vec{\nabla} \times (\vec{\nabla} f)\right)_i.$$

A number equal to its own negative is zero, so every component vanishes and $\vec{\nabla} \times (\vec{\nabla} f) = \vec{0}$. □

B Useful Links

- Laplace relaxation demo:
  https://claude.ai/artifact/J2wPx8V3ajec8vDLsARnKX
- Electrostatics simulation:
  https://icphysweb.z13.web.core.windows.net/simulation.html

[4] This equality uses $\partial_j \partial_i = \partial_i \partial_j$, valid for $\vec{F}$ twice continuously differentiable.